A SMARTER WAY TO RUN A SMALL SHOP

Small shop. Big flow.

Self-service shopping that keeps the customer moving and gives the team a clear view of products, orders, balances and sales.

- Scan-to-cart
- Staff-controlled tabs
- Useful store insights
- Self-hosted

KIOSK APP · RETAIL, MADE REFRESHINGLY PRACTICAL

THE EVERYDAY SHOP CHALLENGE

Less friction at the shelf.
More clarity behind the counter.

A small store still needs a smooth way to browse, buy, keep stock straight and follow up on what is owed.

Make self-service simple

Let customers browse categories, search, build a cart and check out on a shared kiosk.

Turn scans into items

Use common keyboard-style barcode scanners to find products and add them directly to the cart.

Keep tabs visible

See pending balances, order history and payment status in the staff-facing admin panel.

THE SHOPPING FLOW

From beep to basket.

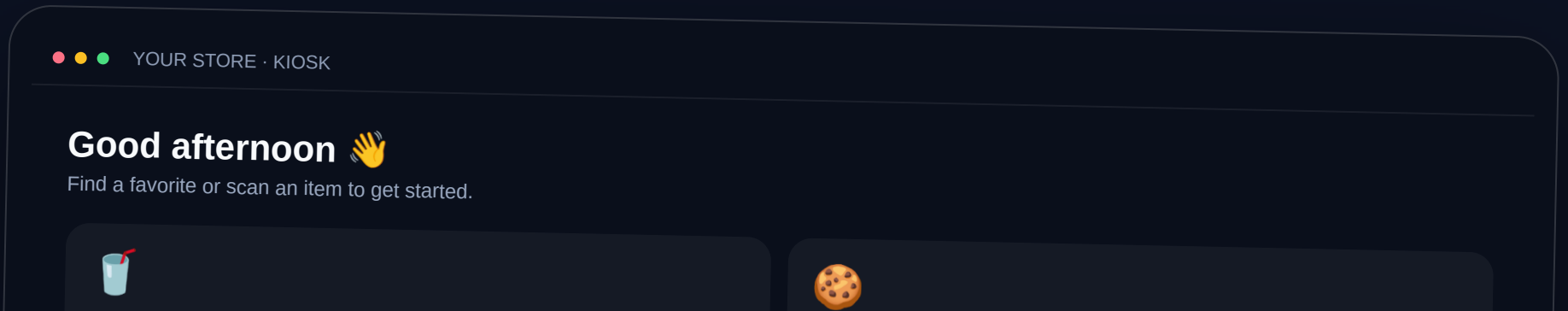
A scanner behaves like a keyboard: scan the code, the kiosk looks up the active product, and the shopper keeps moving.

01 **Staff set it up once**
Attach one or more unique barcode aliases to a product in the admin panel.

02 **Shopper scans**
Scan a barcode into the kiosk field using a USB or Bluetooth HID scanner.

03 **Item lands in the cart**
The matched active product is added; scanning again increases its quantity.

04 **Checkout stays connected**
Stock is checked and the purchase is recorded on the customer's tab.



A BETTER STORE TOOLKIT

Everything essential.
Nothing needlessly complicated.

Customer-first kiosk

Product categories, search, photos, weekly specials, cart totals and customer lookup in one touchscreen-friendly flow.

Flexible barcode setup

Support several barcode aliases for a product; keep lookup exact and shared across product setup and shopping.

Staff stays in charge

Review orders, mark payments, manage stock and control products from the admin panel.

See what is happening

Review customer and product rankings, outstanding balances and sales across day, week and month views.

KNOW THE SHOP, NOT JUST THE CHECKOUT

Turn daily activity into useful answers.

The admin experience brings operational detail into view—without turning the shop into a spreadsheet project.

Top customers

Ranked by paid spend across day, week and month.

Popular products

Most-purchased items and units sold over time.

Payment visibility

Pending balances and recent payment activity at a glance.

Customer context

Order history and useful per-customer statistics.

Insights are based on the data recorded by the store; the app does not claim to forecast demand.

A PRACTICAL OPERATING MODEL

Checkout for a tab-based shop.

Kiosk App records orders for staff to reconcile later. It is not a payment processor or a card terminal.

- **Order recorded**
  Stock is checked and purchase-time item details are saved.
- **Balance stays visible**
  Pending orders contribute to the customer's balance and admin views.
- **Staff closes the loop**
  Mark orders paid, cancel pending orders or reset payment for a customer.

BUILT FOR YOUR OWN ENVIRONMENT

Your shop. Your data. Your deployment.

Run the default app with Docker Compose and a local SQLite file. When traffic calls for it, opt into the PostgreSQL + Redis deployment. The documented runtime does not make outbound network calls; online access is needed to build or obtain container images.

Self-hosted · Docker Compose · SQLite by default

A DEPLOYMENT PATH THAT GROWS WITH THE STORE

Start simple.
Scale when you need to.

The high-traffic setup is an opt-in mode—not a prerequisite for running the kiosk.

DEFAULT · SIMPLE MODE

One store, one host

The existing single-host Compose deployment stays lightweight and straightforward.

- SQLite database file
- One FastAPI service and Nginx frontend
- No Redis or PostgreSQL required

OPTIONAL · HIGH-TRAFFIC MODE

More capacity on the same host

Choose the high-traffic Compose mode when requests or API work justify more replicas.

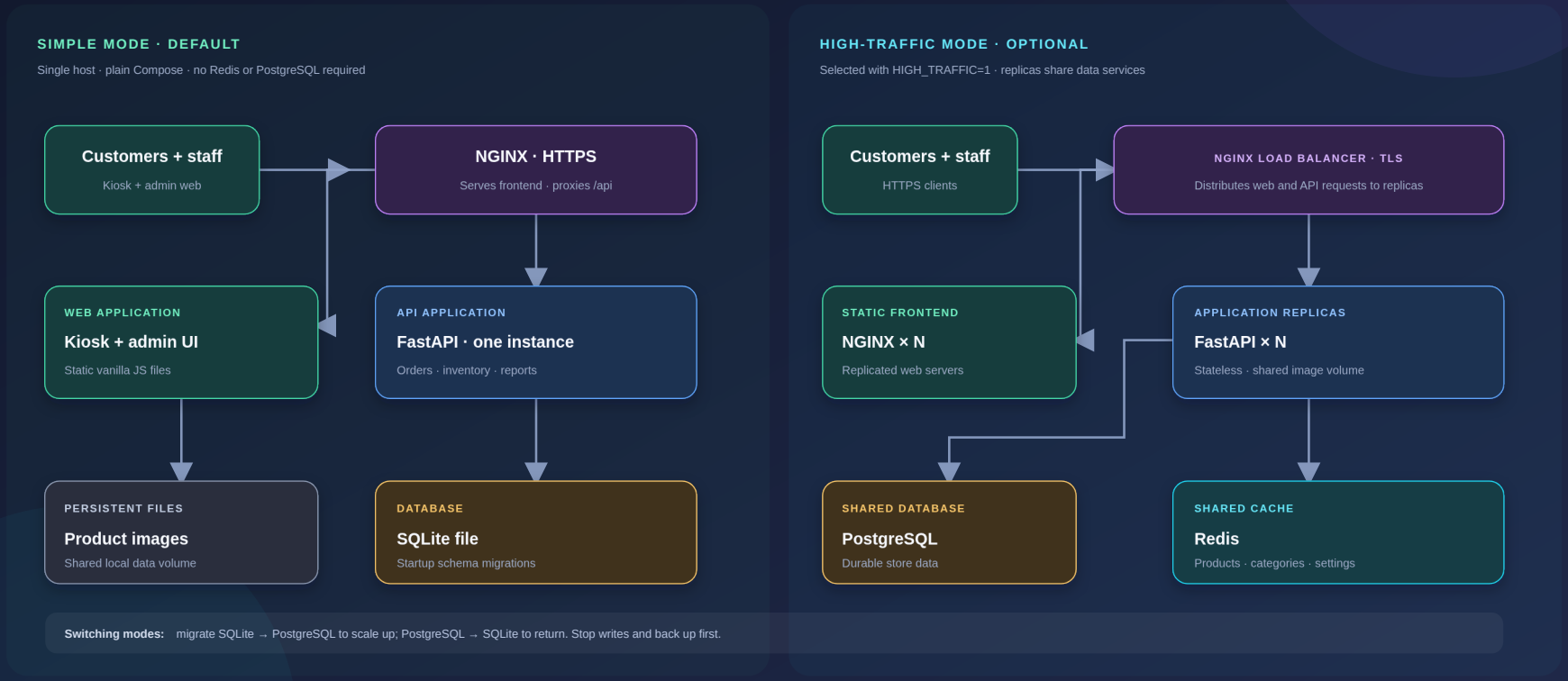
- Nginx load balancer, scalable Nginx and FastAPI replicas
- Shared PostgreSQL database and Redis catalog cache
- Product images available through shared storage

SYSTEM ARCHITECTURE · DEFAULT AND HIGH-TRAFFIC MODES

KIOSK APP · DEPLOYMENT OPTIONS

One shop. Two ways to scale.

Begin with the simple default. Opt into shared services and replicas only when traffic needs them.



The GitHub Pages demo uses an in-browser mock with sample data. High-traffic mode is optional and requires explicit database migration before switching.

YOUR NEXT SHOP SHIFT CAN FEEL SIMPLER

A friendly kiosk for customers.
A clearer picture for staff.

See the interactive demo, explore the code, or bring up your own instance with Docker Compose.

Try the live demo ↗ | Explore the project ↗ | Deployment guide ↗

Kiosk App is a self-hosted store-kiosk project. Features described here are based on the current repository implementation; actual operation depends on local setup and store processes.